

App Development Project Report

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1. Student Details

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About Me: I am a Student at IIT Madras BS Degree program with a deep interest in web application development and data-driven technologies. I enjoy building meaningful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: BS STUDENT PLACEMENT PORTAL

Problem Statement:

To design and build a web-based application that allows users to facilitate the placement process for students, companies, and administrators. Below is an overview of the features and API endpoints available in the portal.

Approach:

The app was built using Flask as the backend framework with sqlalchemy for handling sqllite3 database in application layer. The front end is handle by Vue3 and apis are redis session based.

3. AI/LLM Declaration

I used various AI too to assist in writing the application at various levels like writing hybrid properties in sqlalchemy, using pinia store, designing the pages with css.

The extent of AI/LLM usage is around **15-20%**, limited to **code suggestions and representation layer forming**.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Creating Template for mails
redis	To store task queue, result, caching and session.
celery	To manage the task queue and schedule tasks
Vue3	To handle frontend logic
Vue3-Toastify	To show pop up notifications
Pinia	To Store the state globally in frontend
Vue-Router	To do routing in Frontend
SQLite	Lightweight local database for storing user data

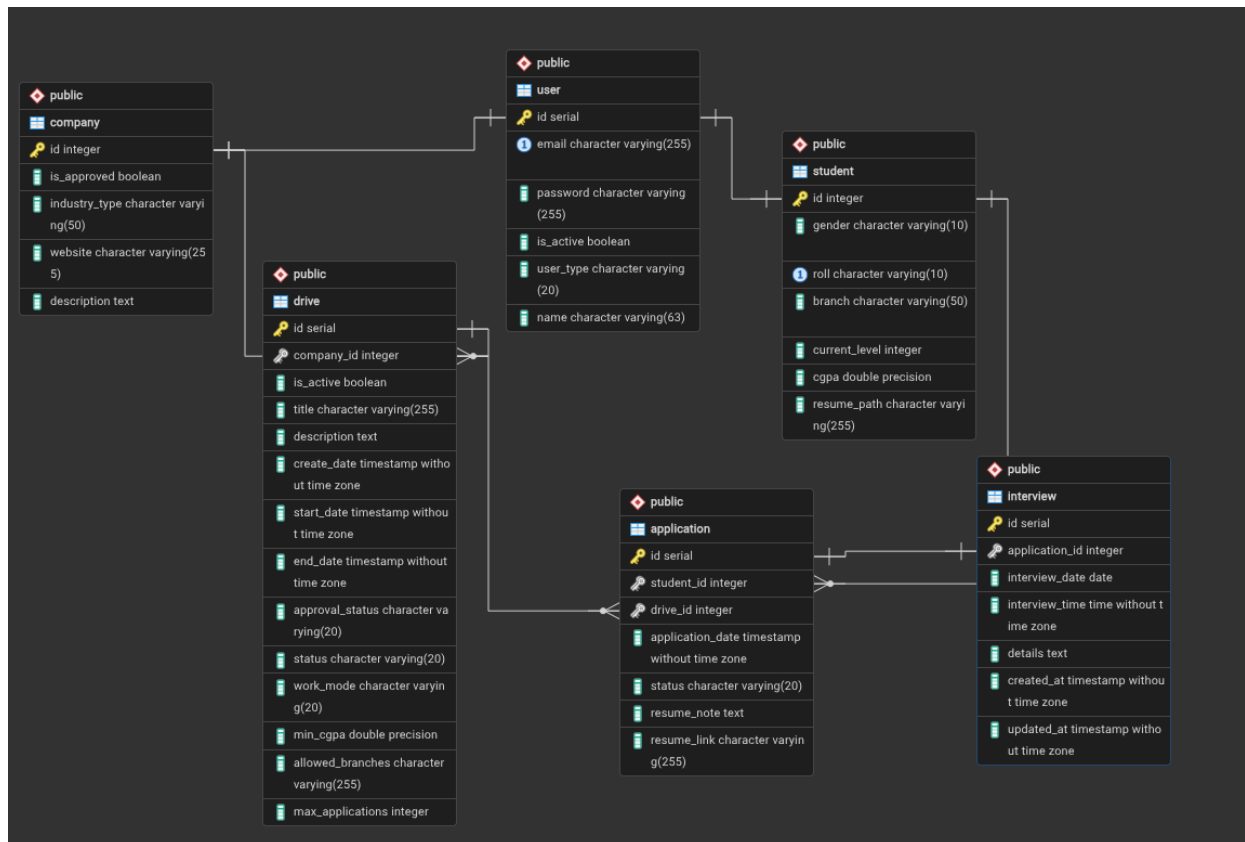
5. Database Schema / ER Diagram

Tables:

1. **User** — stores user profile details (id, name, email, password)
2. **Student** — Stores the information about the Student like(rollnumber, cgpa, branch, etc)
3. **Company** — Stores the information about the company details like (website, industry, etc)
4. **Drives** — Store the information about the drives created by company like(startdate, enddate, etc)
5. **Application** — Stores the information about the student application like (drive_id, studen_id,etc)
6. **Interview** — Stores the information about the interview like time related fields

Relationships:

Below is an er diagram showing the relationship between various entities in detail:



Generated with pgadmin4

6. API Resource Endpoints

Endpoint	Method	Description
/api/	GET	Api Base Route and Landing Api
/api/auth/	GET	Auth Base Route

/api/auth/login/	POST	Login Api for all users
/api/auth/register/company/	POST	Registration Api for Company

YAML API Definition File:

Included a separate yaml file for more details about apis of the application. [link to the file](#)

7. Architecture and Features (optional)

Architecture Overview:

- app.py – main Flask application entry point
- models – database models using SQLAlchemy
- main.js — main entry point in frontend
- router/[index.js](#) – routing logic here
- views/ – frontend views here

Implemented Features:

User registration and login

Daily reminder to students about upcoming drives/interview

Visualization dashboard using Charts

Summary analytics with date filtering

Secure password hashing and user session handling

8. Video Presentation

Drive Link:

[link to video](#)